



State Feedback Control and Observer Design for a 2 DOF Robotic Arm



IE415 Control Of Autonomous System

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Feedback Control Objective

In most applications the objective of a control system is to *regulate or track the system output by adjusting its input subject to physical limitations.* There are two distinct ways by which this objective can be achieved: open-loop and closed-loop.

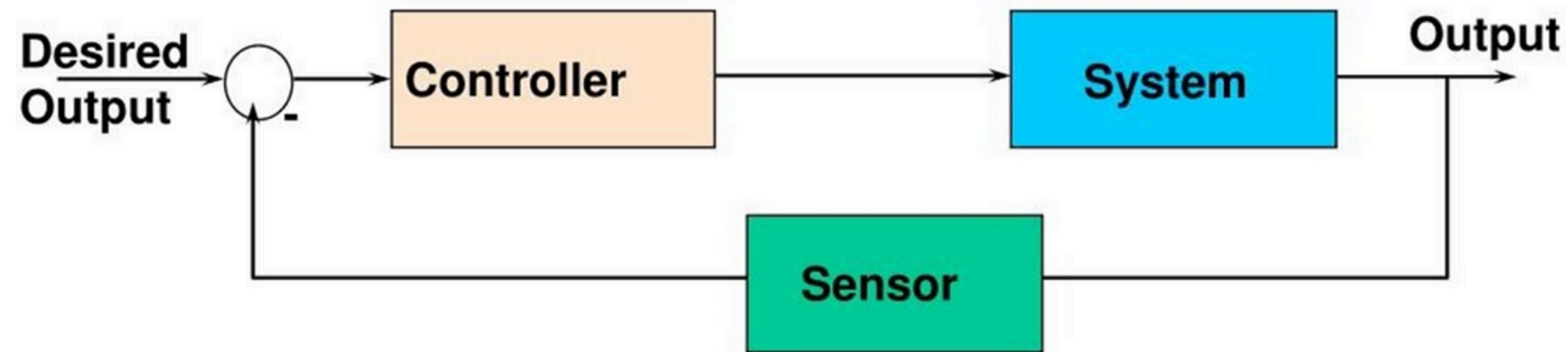


Open-Loop vs. Closed-Loop Control

Open-Loop Control



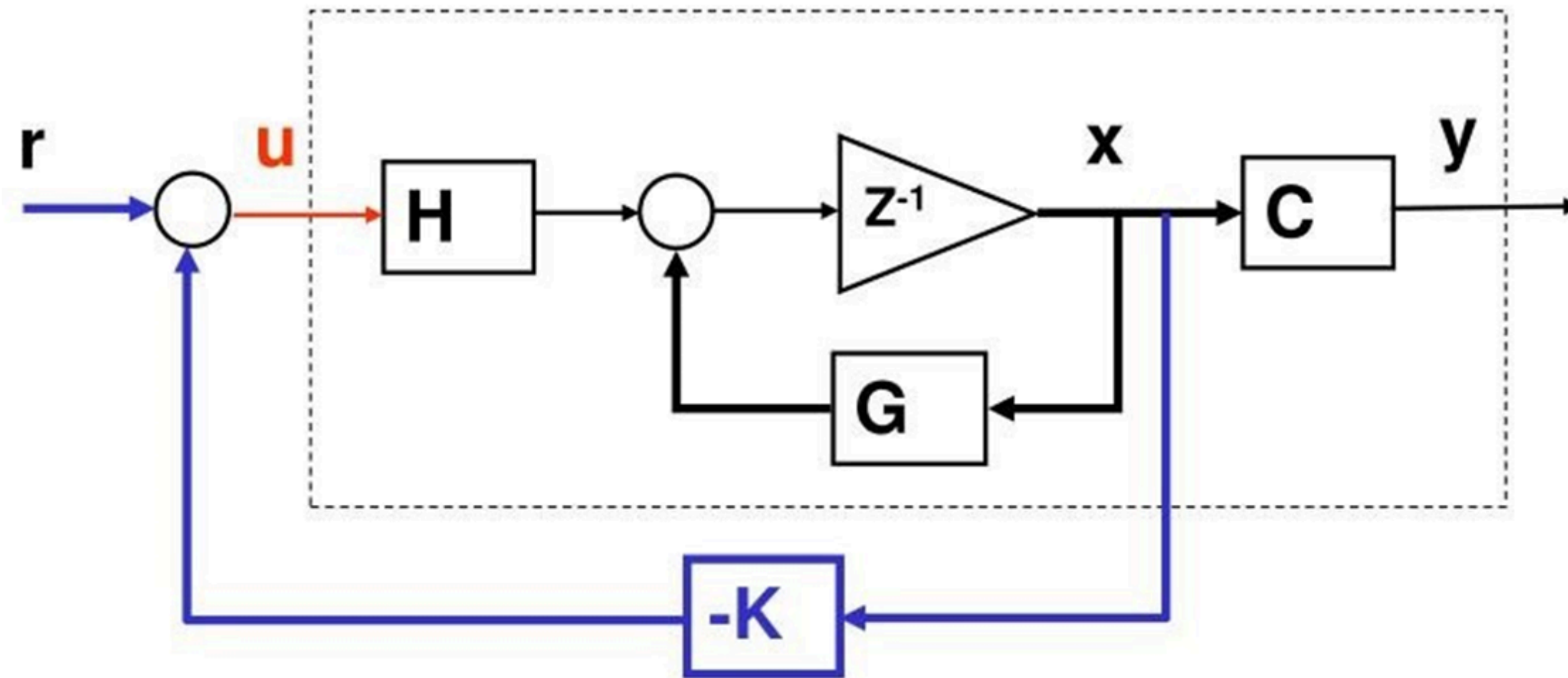
Closed-Loop Control



Closed-Loop Control

- **Advantages**
 - automatically adjusts input
 - less sensitive to system variation and disturbances
 - can stabilize unstable systems
- **Disadvantages**
 - Complexity
 - Instability

State Feedback Control Block Diagram



$$u = -Kx + r$$



State-Feedback Control Objectives

- **Regulation**: Force state x to equilibrium state (usually 0) with a desirable dynamic response.
- **Tracking**: Force the output of the system y to track a given desired output y_d with a desirable dynamic response.



Closed-loop System

Plant: $\mathbf{x}(k+1) = \mathbf{G}\mathbf{x}(k) + \mathbf{H}\mathbf{u}(k)$
 $\mathbf{y}(k) = \mathbf{C}\mathbf{x}(k)$

Control: $\mathbf{u} = -\mathbf{K}\mathbf{x} + \mathbf{r}$

Closed-loop System:

$$\mathbf{x}(k+1) = (\mathbf{G} - \mathbf{H}\mathbf{K})\mathbf{x}(k) + \mathbf{H}\mathbf{r}(k)$$
$$\mathbf{y} = \mathbf{C}\mathbf{x}$$

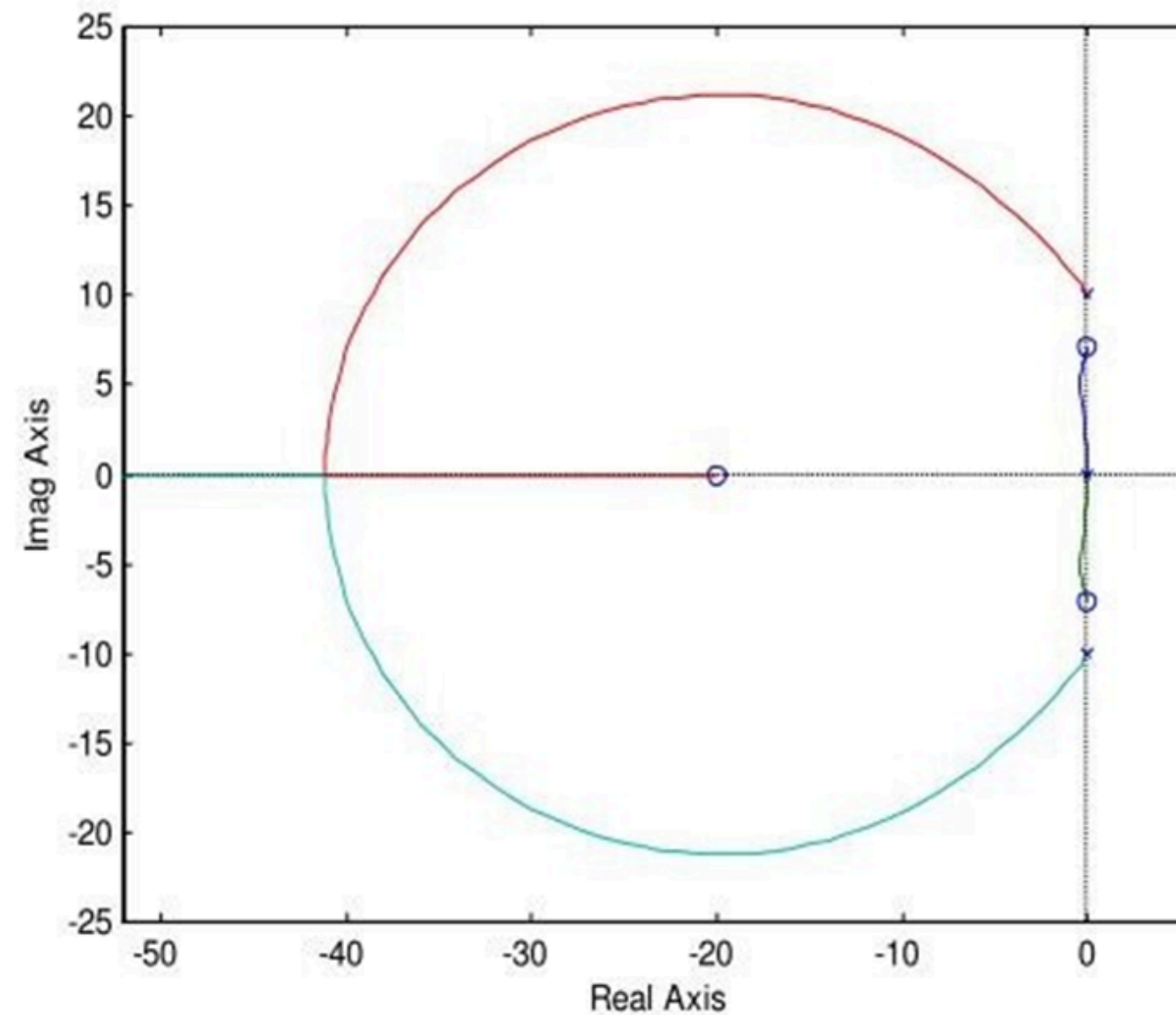


Collocated Control

Transfer Function: $G_p = \frac{X_1(s)}{U(s)} = \frac{s^2 + 50}{s^2(s^2 + 100)}$

PD Control: $G_c = K(s + a), \quad a = 20$

Root-
Locus





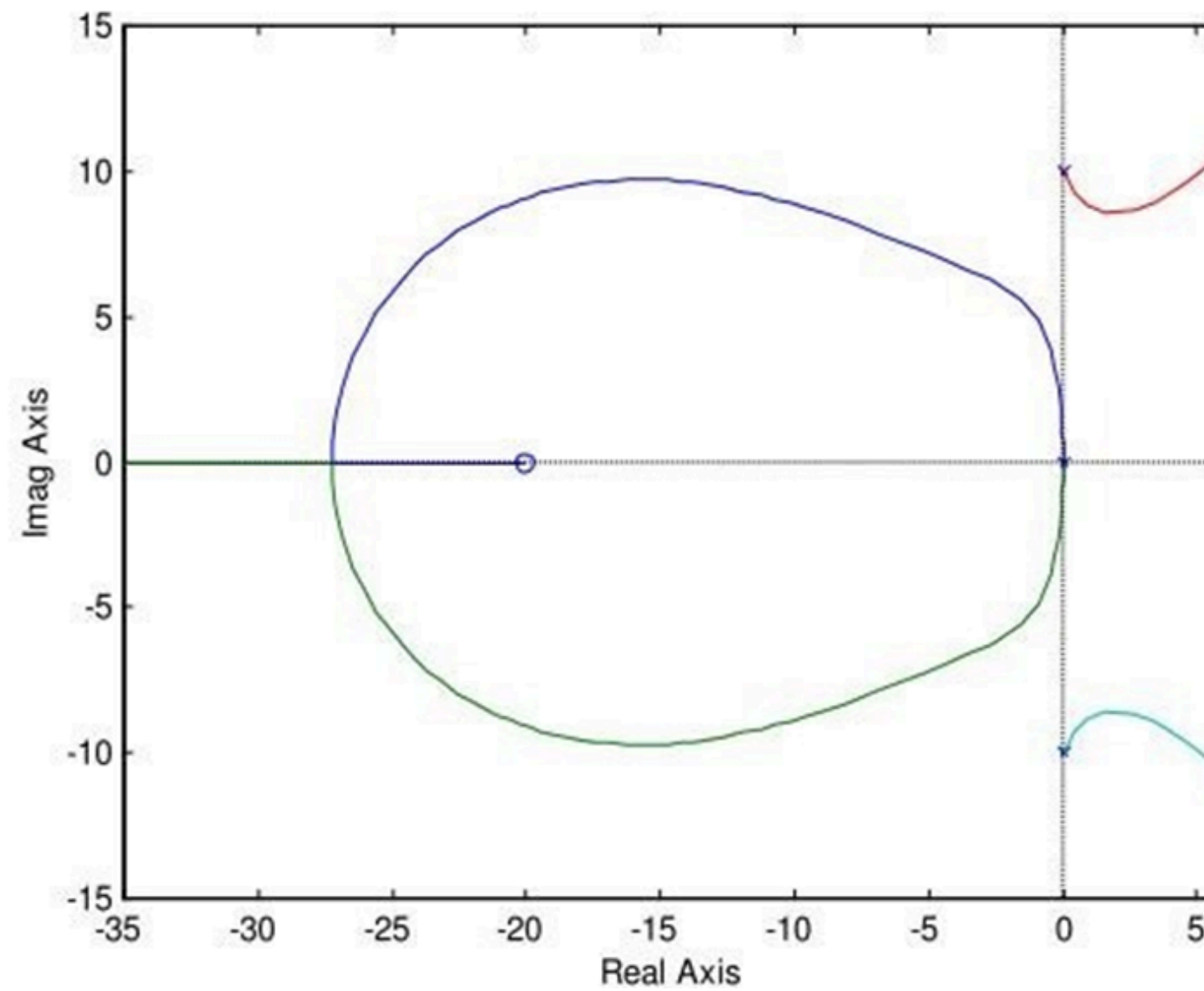
Non-Collocated Control

Transfer Function: $G_p = \frac{X_2(s)}{U(s)} = \frac{50}{s^2(s^2 + 100)}$

PD Control: $G_c = K(s + a), \quad a = 20$

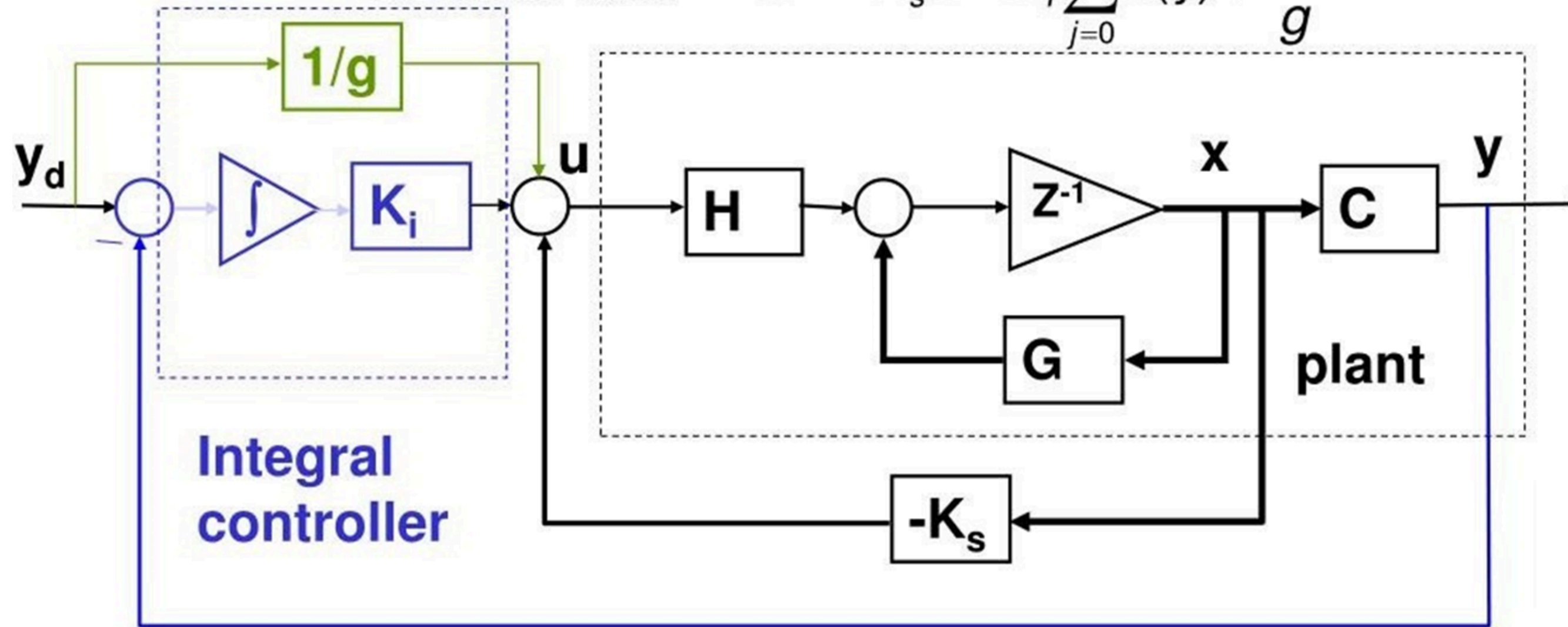
Root-
Locus

Unstable



Integral Control

Control law: $u = -K_s x - K_i \sum_{j=0}^{k-1} e(j) + \frac{y_d}{g}$



Automatically generates reference input r !



Degree Of Freedom (DOF)

(a) **Degree of freedom of a point in space:**

- Degree of freedom (DOF) of a point is the number of independent coordinates required to define the position and orientation of a point.
- Thus a point in space has only three degrees of freedom or three coordinates are required to define its position. The point P can be uniquely defined by specifying its three co-ordinates x , y and z .

(b) **Degree of freedom (DOF) of a rigid body :**

- Consider a rigid body moving in space as shown in Fig. 1.34. This rigid body have possesses following independent motion.
 - 1) Translation motions along any three perpendicular axis x , y and z .



2) Three rotational motions about the axis x , y and z . Thus degree of freedom of rigid body have six degree of freedom in free space.

(c) Degree of freedom (DOF) of a mechanism :

➤ A mechanism can also have several degree of freedom (DOF).

> The degree of freedom (DOF) of a mechanism is decided by the degree of freedom (DOF) of the links constituting that mechanism. Accordingly, we can classify mechanism under two general categories as follows:

- 1) Spatial mechanism.
- 2) Planner mechanism.

2 DOF Manipulator

Now, given the joint angles θ_1 , θ_2 we can determine the end effector coordinates x and y .

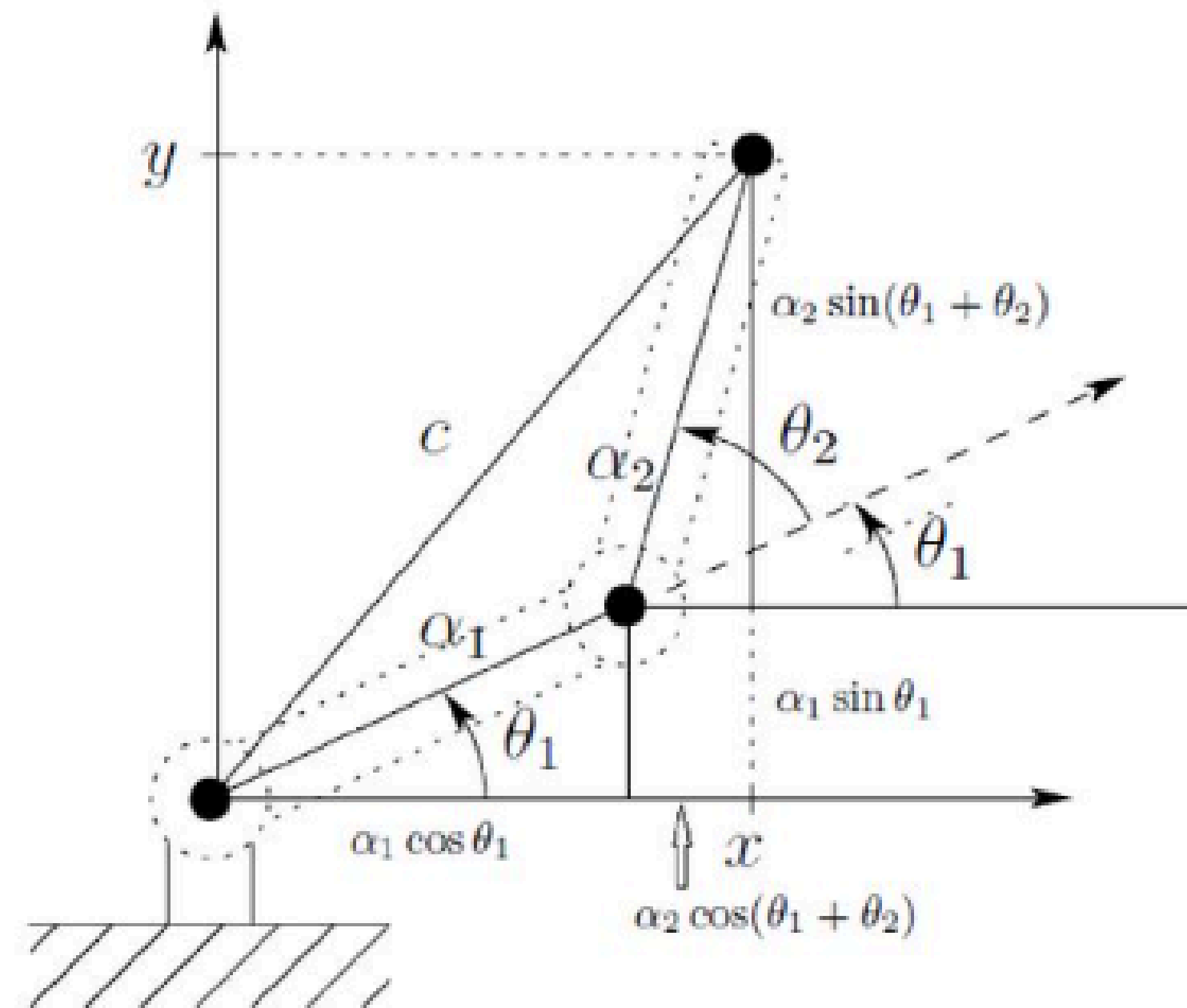


Fig. 1.26 Solving for the joint angles of a two-link planar arm.



Forward kinematics

The coordinates (x, y) of the tool are expressed in this coordinate frame as:

$$x = x_2 = \alpha_1 \cos \theta_1 + \alpha_2 \cos(\theta_1 + \theta_2) \quad (1.1)$$

$$y = y_2 = \alpha_1 \sin \theta_1 + \alpha_2 \sin(\theta_1 + \theta_2) \quad (1.2)$$

in which α_1 and α_2 are the lengths of the two links, respectively. Also the **orientation of the tool frame** relative to the base frame is given by the direction cosines of the x_2 and y_2 axes relative to the x_0 and y_0 axes, that is,

$$x_2 \cdot x_0 = \cos(\theta_1 + \theta_2); \quad x_2 \cdot y_0 = -\sin(\theta_1 + \theta_2)$$

$$y_2 \cdot x_0 = \sin(\theta_1 + \theta_2); \quad y_2 \cdot y_0 = \cos(\theta_1 + \theta_2)$$

which we may combine into an orientation matrix

$$\begin{bmatrix} x_2 \cdot x_0 & y_2 \cdot x_0 \\ x_2 \cdot y_0 & y_2 \cdot y_0 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{bmatrix} \quad (1.3)$$



Inverse Kinematics

In order to command the robot to move to a location we need the inverse; that is, we need the joint variables θ_1 , θ_2 in terms of the x and y coordinates of A.

This is the problem of inverse kinematics. In other words, given x and y in the forward kinematic Equations (1.1) and (1.2).

Consider the diagram of Figure 1.26. Using the Law of Cosines we see that the angle θ_2 is given by

$$\cos \theta_2 = \frac{x^2 + y^2 - \alpha_1^2 - \alpha_2^2}{2\alpha_1\alpha_2} := D \quad (1.4)$$

We could now determine θ_2 as

$$\theta_2 = \cos^{-1}(D) \quad (1.5)$$

However, a better way to find θ_2 is to notice that if $\cos(\theta_2)$ is given by Equation (1.4) then $\sin(\theta_2)$ is given as

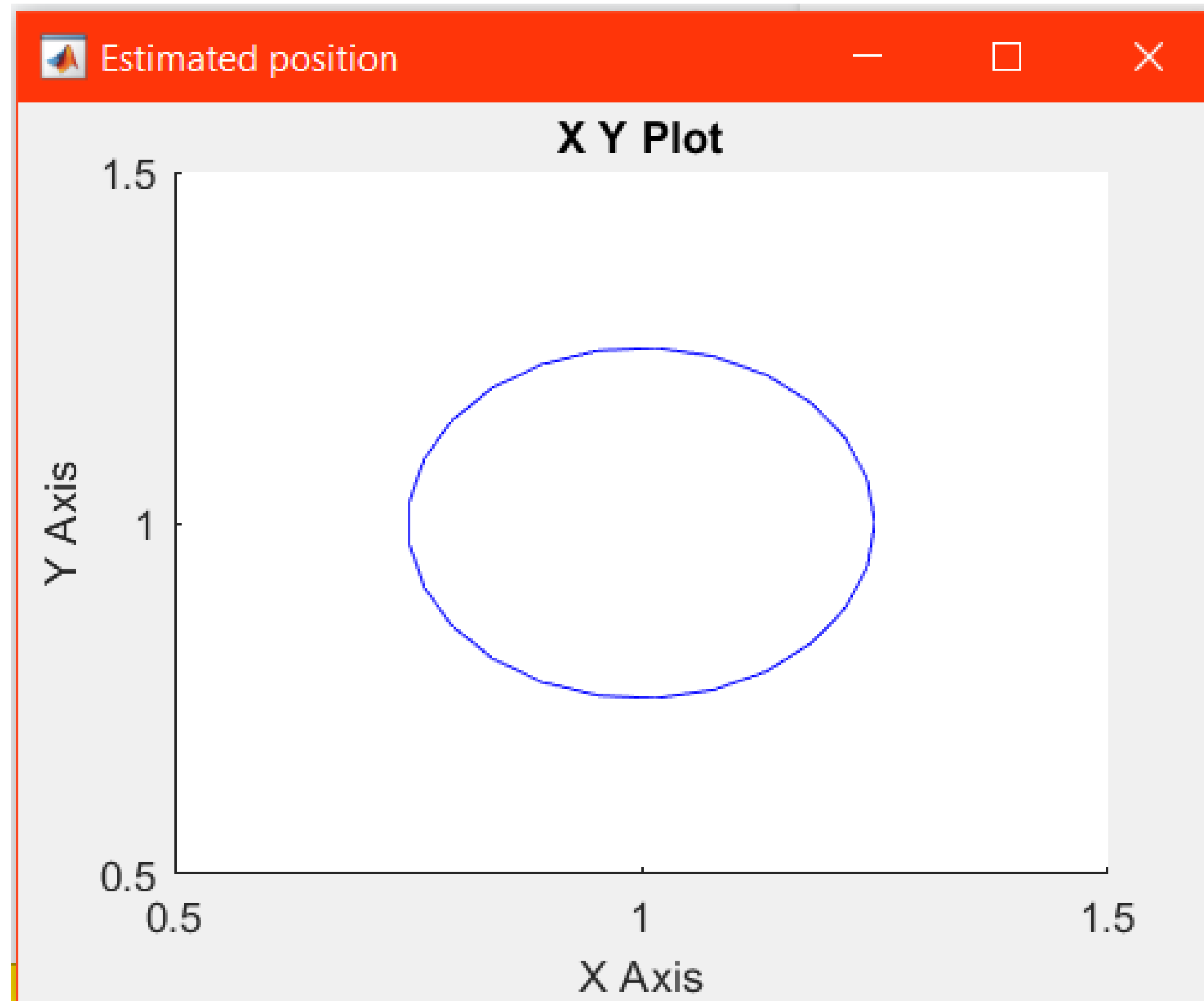
$$\sin(\theta_2) = \pm\sqrt{1 - D^2} \quad (1.6)$$

and, hence, θ_2 can be found by

$$\theta_2 = \tan^{-1} \frac{\pm\sqrt{1 - D^2}}{D} \quad (1.7)$$



Robo arm 2D Trajectory



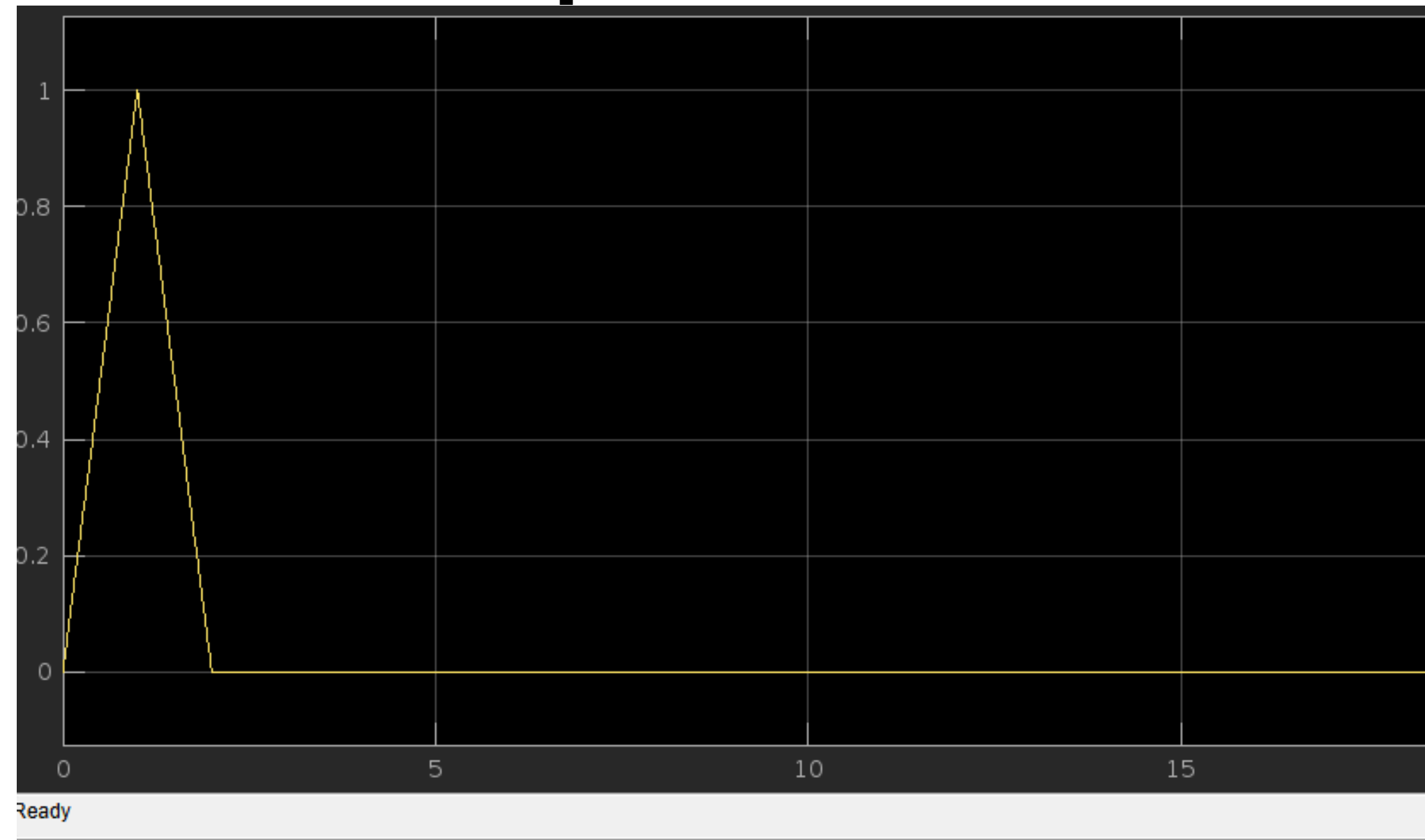


Robo arm Simulink Result





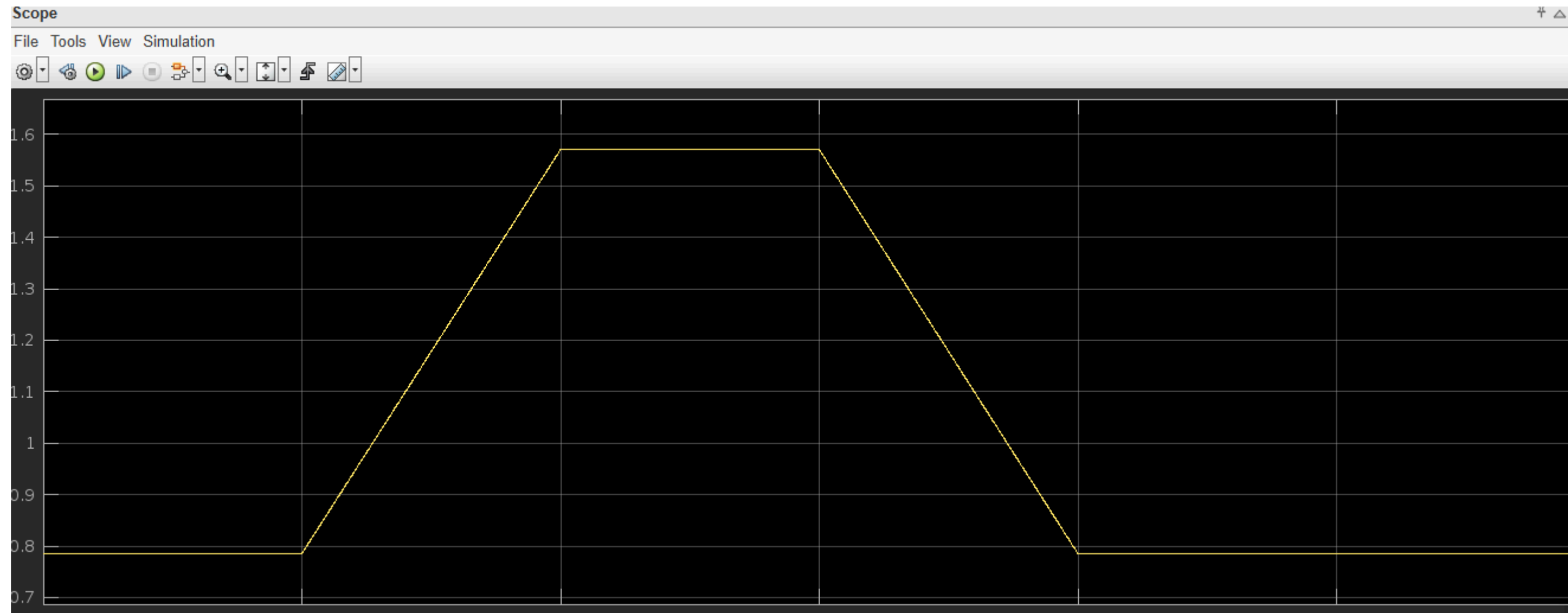
2 DOF Simscape Matlab Results



This shows a signal that spikes initially and then settles to zero, which represent an impulse response or a damped system reaching equilibrium (possibly acceleration or force on one DOF).



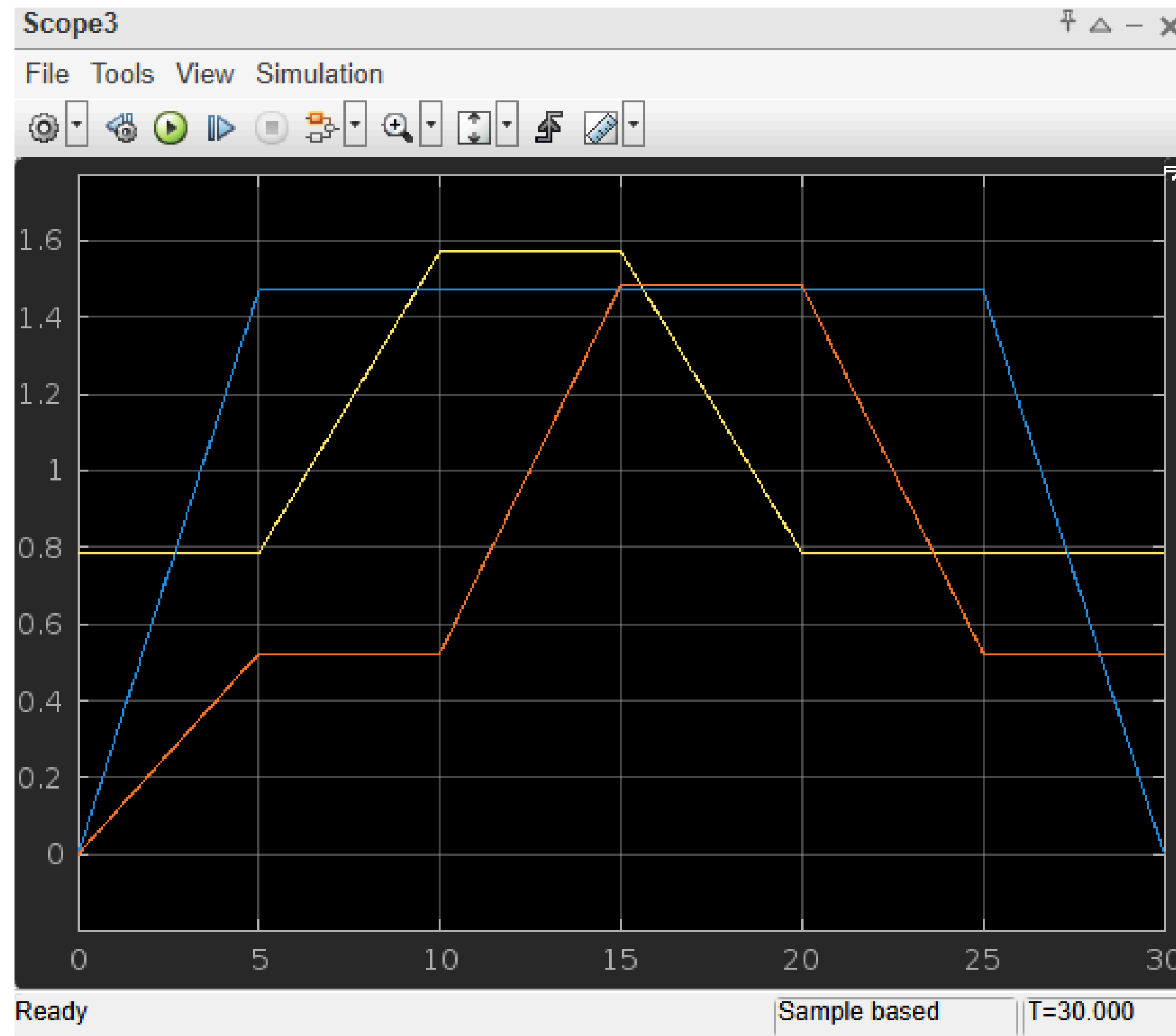
2 DOF Simscape Matlab Results



This graph shows a single signal rising, holding a plateau, and then decreasing symmetrically, which is characteristic of a displacement or position signal that follows a trajectory over time.



2 DOF Simscape Matlab Results





Thank You